

Mohit Agrawal

agrawalmohit.ma@gmail.com | 901-124-4748

Location: Nagpur, India (Willing to relocate)

PROFESSIONAL SUMMARY

AI Engineer with 3+ years of experience in **data analytics, machine learning, forecasting, and AI application development**. Experienced in **Python, SQL, predictive modeling, NLP, and LLM-based** applications. Built end-to-end AI projects using **LangChain, OpenAI GPT, FastAPI, and vector databases**, with professional experience delivering analytics, forecasting, and business intelligence solutions. **Passionate about building production-ready AI systems, LLM-powered applications, and scalable machine learning solutions.**

EDUCATION AND HONORS

International University of Applied Sciences, Berlin, Germany

Master Of Science (M.S.) in Computer Science

Sant Gadge Baba Amravati University, Amravati, Maharashtra

Bachelor of Engineering (B.E.) in Computer Science and Engineering

[July 2018-July 2022]

TECHNICAL SKILLS

Programming Languages : Python, SQL, Java, C++

Machine Learning : Regression, Classification, Clustering, Time Series Forecasting, Feature Engineering, Model Evaluation, Hyperparameter Tuning

Deep Learning : TensorFlow, PyTorch, Neural Networks, LSTM, Transformers

Natural Language Processing & Generative AI : OpenAI GPT, LangChain, Hugging Face, BERT, spaCy, Retrieval-Augmented Generation (RAG), Prompt Engineering, Embeddings, FAISS, Vector Databases, Agentic AI, Tool Calling

Frameworks & Deployment : FastAPI, Flask, Streamlit, Docker, Git, REST APIs

Data Analytics & Visualization : Power BI, Tableau, Excel

Databases : PostgreSQL, MySQL

Agentic AI: ReAct, Tool Calling, MCP (Model Context Protocol)

PROFESSIONAL EXPERIENCE

AI/ML & BI Developer

Highbar Technocrat (Client : Maharashtra Metro Rail Corporation Ltd.) – Nagpur

March 2025 – Present

- Developed and maintained 10+ operational dashboards for **AFC, OCC, CRM, revenue, ridership, and station performance using Power BI.**
- Automated daily **data ingestion, validation, and reporting workflows** using SQL and Python.
- Collaborated with cross-functional teams to deliver **forecasting models for passenger demand and revenue planning.**
- Built **anomaly detection logic** to identify unusual **passenger flow** and operational deviations.

- Designed **KPI dashboards** for senior management supporting operational and **business decisions**
 - Collaborated with business stakeholders to gather requirements and translate operational challenges into **analytical and forecasting solutions**.
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Database Operations Analyst
Bamboobox, Bangalore (A product-based company)
January 2025 – March 2025

- Conducted **data enrichment and data mining** using tools like **LSN, Apollo, and Lusha**.
 - Cleaned, processed, and validated **large datasets** to improve **lead generation and client prospecting**.
 - Optimized **data pipelines** to improve data quality and usability.
 - Ensured **data integrity, quality and compliance and governance standards**.
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Data Science & Machine Learning Engineer
Samatrix Consulting Pvt. Ltd., Gurgaon
July 2024 – December 2024

- Built **time-series and regression-based forecasting models** to predict business trends.
 - Cleaned and processed large multi-source datasets using **Python (Pandas, NumPy)**.
 - Applied **feature engineering techniques** to improve forecast reliability and stability.
 - Built **Power BI and Tableau dashboards** to communicate forecasting insights.
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Data Analyst / Machine Learning Engineer
Starlink Technology Pvt. Ltd., Pune
March 2023 – July 2024

- Built **predictive and demand forecasting models** using Python, SQL, and scikit-learn.
 - Conducted **EDA, feature engineering, and hypothesis testing** to improve forecast accuracy.
 - Developed **trend and time-series models** using large historical datasets.
 - Created automated **Tableau dashboards** to track trends and forecast outputs.
 - Supported **data-driven planning** through reliable forecasting insights..
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PROJECTS

AI Resume Intelligence System (ATS + Semantic Matching)

Tech Stack: Python, FastAPI, Sentence Transformers (BERT), spaCy, scikit-learn, PDFMiner, Docker, Hugging Face

- Built an **AI-powered Resume Intelligence System** to parse **PDF resumes**, calculate **ATS scores**, and perform **semantic job matching** using **NLP and transformer embeddings**.
- Developed **resume parsing modules** to extract candidate information **including contact details, skills, education, and experience** from unstructured PDF resumes.
- Implemented **BERT-based semantic similarity** using **Sentence Transformers** to compare resumes with **job descriptions and generate match scores**.

- Designed an **ATS scoring engine** to evaluate resume-job fit **and identify matched and missing skills**.
- Exposed functionality through **FastAPI REST APIs** and generated structured **JSON responses** for **downstream recruitment** workflows.
- Containerized the application **using Docker** and **deployed it on Hugging Face Spaces** for consistent and scalable execution.

Retrieval-Augmented Generation (RAG) based PDF Question Answering System

Tech Stack: Python, FastAPI, Streamlit, LangChain, Hugging Face, FAISS, Docker

- Built an end-to-end **Retrieval-Augmented Generation (RAG) application** to **answer natural language questions** from PDF documents.
- Developed a document ingestion pipeline using **LangChain, PyPDFLoader, and Recursive Character Text Splitter** for semantic document processing.
- Generated **semantic embeddings using Hugging Face Sentence Transformers** and **indexed documents in FAISS** for efficient vector similarity search.
- Implemented **context-aware retrieval** to fetch the most relevant document chunks before **generating grounded AI responses**.
- Designed prompt templates to improve **response relevance** and **reduce hallucinations**.
- Developed **REST APIs using FastAPI** and an **interactive Streamlit frontend** for document upload and **question answering**.
- **Containerized the application using Docker** and structured the project for modular, production-style development.

AI Inventory Management Agent

Tech Stack: Python, FastAPI, React, LangChain, Groq (Llama 3.3 70B), PostgreSQL, SQLAlchemy, Docker

- Built an end-to-end **AI Inventory Management Agent** capable of interpreting **natural language requests** and executing **multi-step inventory workflows**.
- Implemented ReAct-style reasoning with tool calling for **inventory search, low-stock detection, order history retrieval, order creation, status updates**, and order cancellation.
- Integrated **PostgreSQL with SQLAlchemy** for inventory and order management and exposed **REST APIs using FastAPI**.
- **Implemented input validation** and structured exception handling **to improve API reliability** and user experience.
- Containerized the **frontend and backend using Docker** and Docker Compose for consistent local deployment.

CERTIFICATIONS

- **Google Data Analytics** Specialization (Google)
- **Data Analytics and Visualization** Job Simulation (Accenture North America)
- **Data Analytics Consulting** Virtual Internship (KPMG)
- **Artificial Intelligence** Virtual Experience Internship (Cognizant)

ACHIEVEMENTS

- Built multiple AI and ML applications using **LangChain, FastAPI and OpenAI APIs**.
- **Developed production dashboards** used for operational monitoring.
- Active **Github contributor** with projects in **Gen AI and Machine Learning**

- Developed **AI powered automation and analytics solutions** for business reporting and decision support

PROFILE LINK

[LinkedIn](https://www.linkedin.com/in/mohit-agrawal45) :- **<https://www.linkedin.com/in/mohit-agrawal45>**

[GitHub](https://github.com/Mohhitt45) :- **<https://github.com/Mohhitt45>**
